

Shashank Mankala

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SUMMARY

AI Engineer / Applied Machine Learning Engineer with experience building production-grade LLM systems, inference engines, and end-to-end ML pipelines. Strong background in GenAI architectures, decoding strategies, RAG systems, and scalable ML infrastructure across cloud environments.

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript, TypeScript, SQL

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, AWS Bedrock, vLLM, Vector Databases, Embeddings, Model Evaluation, LLM Serving, Inference Optimization

Backend Development: FastAPI, Flask, Node.js, REST APIs, API Development, WebSockets

Cloud & Databases: AWS, Azure, GCP, PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

Data Engineering & DevOps: ETL, Workflow Orchestration, Airflow, Snowflake, Databricks, Docker, Kubernetes, GitHub Actions, CI/CD

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, System Design, Design Patterns, Distributed Systems, Concurrent Programming, Asynchronous Processing, Service-Oriented Architecture, Linux, Git

Certification: Microsoft Certified: ML Operations Engineer Associate, SQL AI Developer Associate

PROFESSIONAL EXPERIENCE

AI Engineer – Tiger Analytics, Santa Clara, CA

Feb 2026 – Present

- Engineered an AI-powered garment **cost estimation** platform that automated first-cost prediction from unstructured design inputs by using Gemini-powered attribute extraction, product classification, and CatBoost pipelines through FastAPI.
- Designed garment classification and feature extraction pipelines that transformed visual product sketches into structured machine learning datasets, enabling scalable and automated downstream cost prediction workflows.
- Built a similarity retrieval system using **FashionCLIP embeddings** and vector search to surface the top 3 visually related garments for cost validation and explainability, improving transparency and stakeholder confidence in model predictions.
- Deployed and scaled containerized backend and model-serving applications on **Google Cloud Platform**, leveraging GKE, Cloud Run, Vertex AI, Artifact Registry, and CI/CD pipelines to support reliable production inference workflows.

Data Scientist Intern – New Era Cap, Buffalo, NY

Sep 2025 – Dec 2025

- Architected an end-to-end **demand forecasting** platform supporting 40K+ SKUs by building Snowflake-based data engineering pipelines, feature generation workflows, and automated model retraining processes to enable scalable forecasting operations.
- Developed and deployed a production forecasting application on **GCP** that generated quarterly, semi-annual, and annual demand projections, while implementing monitoring and validation frameworks to improve forecast reliability.

Lead Data Analyst – Shadowfax, Bangalore, India

Sep 2023 – Jul 2024

- Engineered scalable data pipelines and analytical workflows to optimize logistics routing and workforce allocation, enabling data-driven decision-making across high-volume delivery operations.
- Developed operational analytics and performance monitoring frameworks in collaboration with cross-functional stakeholders, improving visibility into workforce productivity, delivery performance, and resource utilization.

Software Engineer – Frugal Testing, Hyderabad, India

Jun 2022 – Jun 2023

- Developed application modules and backend workflows across EdTech, healthcare, and fintech platforms, contributing to user-facing features, business process automation, and scalable software delivery while building reusable frameworks.
- Contributed to a Central Bank Digital Currency (CBDC) platform in collaboration with **NPCI**, implementing API routing and service integration workflows within a **blockchain** ecosystem and production readiness through automated validations.

PROJECTS

AUTOapply (AI-Powered Automation Platform)

- Architected and developed a distributed automation platform supporting Greenhouse, Lever, and Ashby ATS workflows through modular service design, dynamic ATS routing, browser automation, and fault-tolerant execution pipelines.
- Built a multi-provider LLM orchestration layer integrating Groq, OpenRouter, and Ollama with automated fallback, structured information extraction, and human-like interactions using Playwright, stealth techniques, and behavioral simulation.

High-Performance LLM Inference and Agentic AI Engine

- Engineered a LLM serving platform implementing streaming inference, KV caching, continuous batching, configurable decoding strategies (greedy, temperature, top-k/top-p sampling), and OpenAI-compatible APIs to optimize throughput.
- Developed an agentic AI framework with workflow execution and benchmarking capabilities, evaluating Hugging Face and vLLM backends using latency, throughput, and time-to-first-token (TTFT) metrics.

EDUCATION

State University of New York at Buffalo

Aug 2024 - Dec 2025

Master's in Data Science, Buffalo, United States

Lovely Professional University, Punjab

Aug 2019 - May 2023

Bachelor's in Computer Science and Engineering